

Logistic Regression

Definition

→ logistic regression is a method used for classification problems.

→ In this, we find probability that a given input belongs to a particular category.

→ In linear regression, we find

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \dots + \theta_n x_m$$

but it gives values in range

$$[-\infty, \infty]$$

→ we use sigmoid function that converts a real-value in $[-\infty, \infty]$ into real-value in

$[0, 1]$

what is Sigmoid Function?

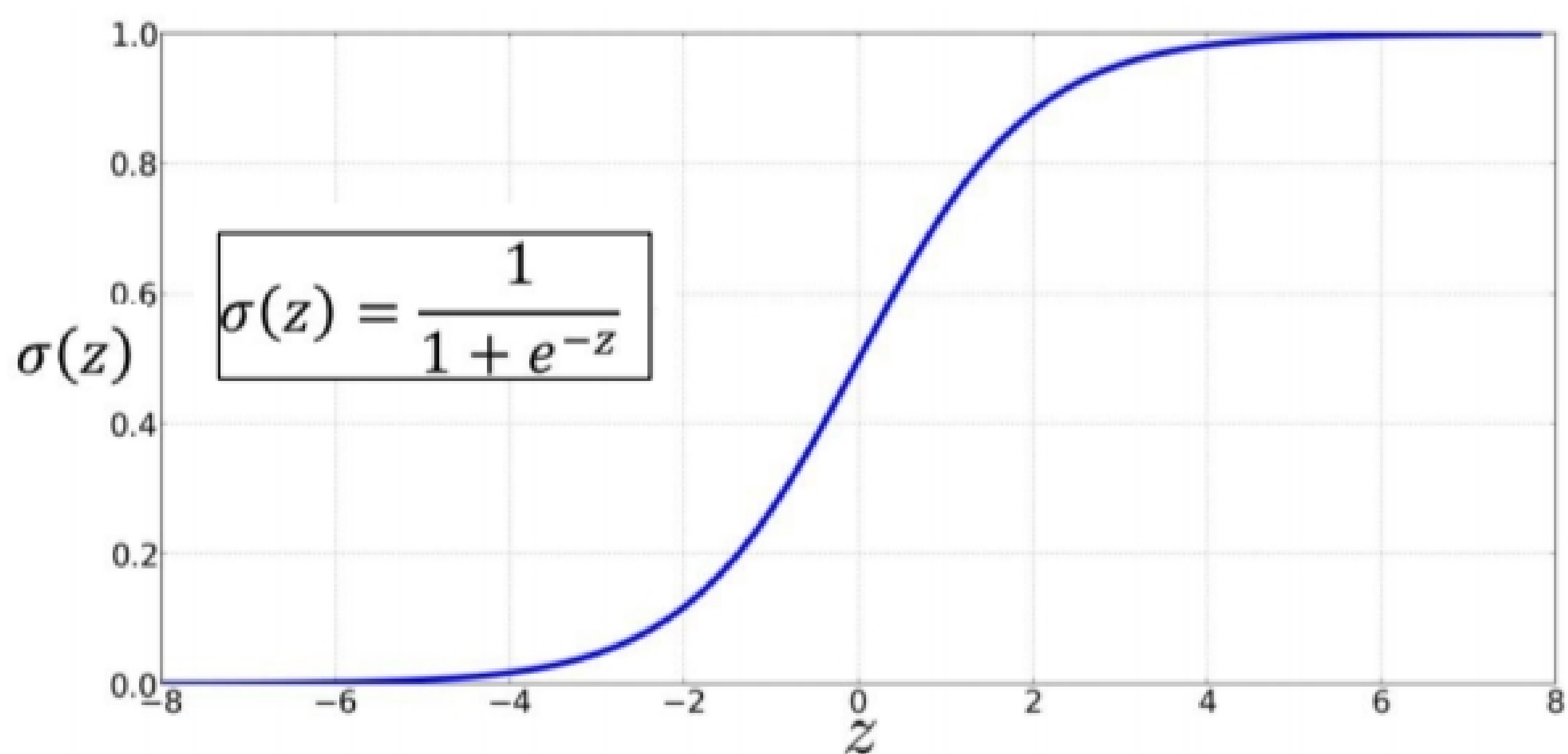
$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

→ If x is +ve, it will make it -ve, hence denominator $(1 + e^{-x})$ would become low, hence giving value more close to 1.

→ If x is -ve, denominator $(1 + e^x)$ would become high, thus giving value more close to 0.

why is it called "sigmoid"?

→ when we plot $\sigma(x)$ for values of x in $[-\infty, \infty]$, it gives function curve of shape "S".



→ so we take sigmoid of linear function $h_\theta(x)$ to get probabilities in range $[0, 1]$

$$\sigma(h_\theta(x)) = \frac{1}{1 + e^{-h_\theta(x)}}$$

Derivation of $\sigma(h_0(x))$

→ we model Odds to derive equation of logistic regression.

→ Odds are defined as:
ratio of occurring of an event with its non-occurring

$$\text{odds} = \frac{p(x)}{1 - p(x)}$$

→ we assume that odds grow exponentially with linear score:

$$\text{linear score} = z = wX + b$$

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}, \quad b = b_0$$

$$\frac{p(x)}{1-p(x)} = e^z$$

If we take log on both sides

$$\log \left[\frac{p(x)}{1-p(x)} \right] = \log e^z$$

✓
called log-odds
or logit

just for understanding

$$\frac{p(x)}{1-p(x)} = e^{wx+b}$$

$$p(x) = e^{wx+b} (1-p(x))$$

$$= e^{wx+b} - e^{wx+b} \times p(x)$$

$$p(x) + e^{wx+b} \times p(x) = e^{wx+b}$$

$$p(x) (1 + e^{\omega x + b}) = e^{\omega x + b}$$

$$p(x) = \frac{e^{\omega x + b}}{1 + e^{\omega x + b}}$$

Dividing denominator and numerator by $e^{\omega x + b}$

$$\Rightarrow p(x) = \frac{e^{\omega x + b}}{1 + e^{\omega x + b}} \bigg/ e^{\omega x + b}$$

$$(1 + e^{\omega x + b}) \bigg/ e^{\omega x + b}$$

$$= \frac{1}{\frac{1}{e^{\omega x + b}} + 1} = \frac{1}{1 + e^{-(\omega x + b)}}$$

$$\therefore p(x) = P(y=1/x)$$

- there is huge difference b/w

$p(x)$ and $P(x)$.

- $p(x)$ is short way of writing

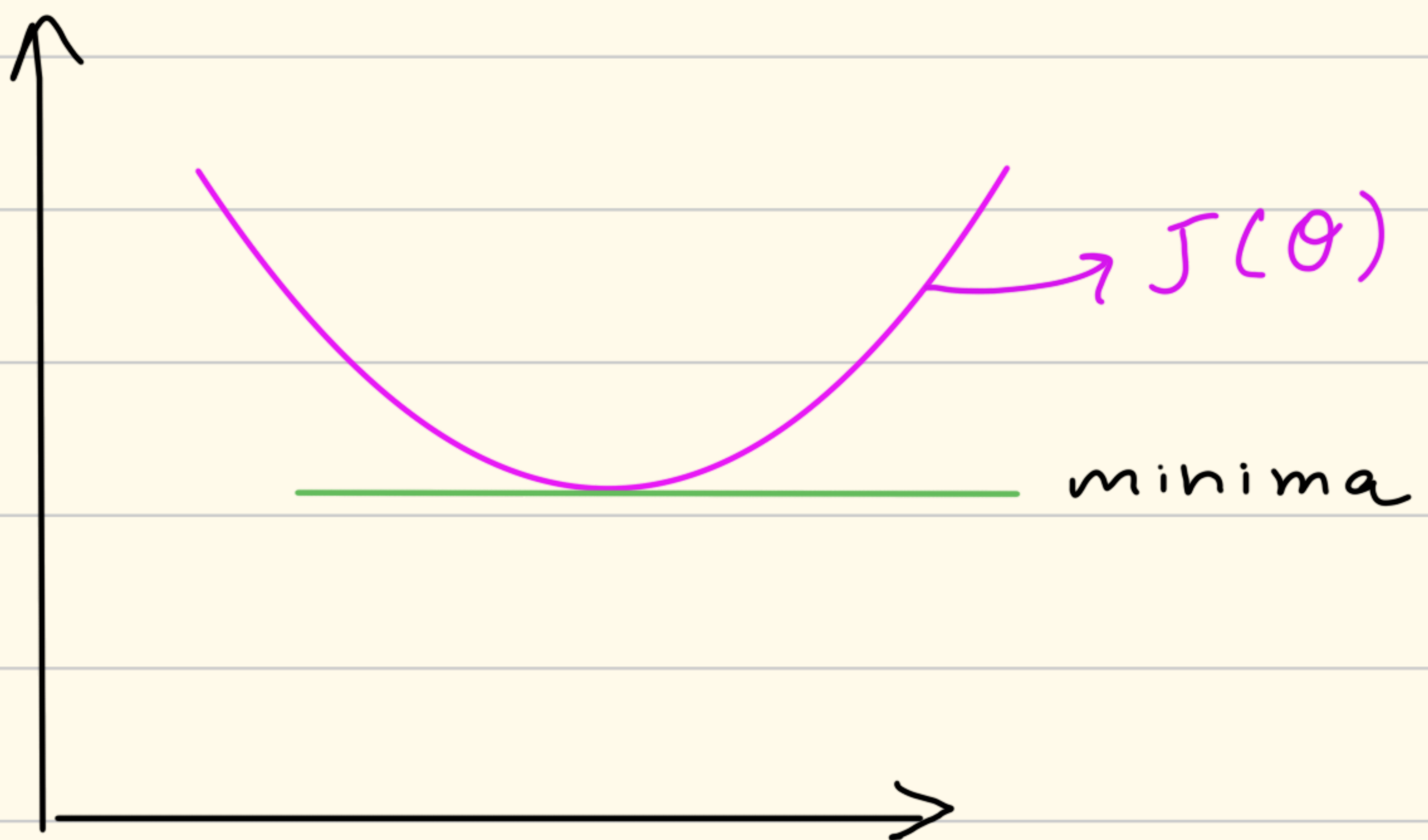
$$P(y=1/x)$$

How do we choose ω ?

→ In linear regression, we used cost function

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2$$

- we minimized it and got θ (or ω)
- it was like this

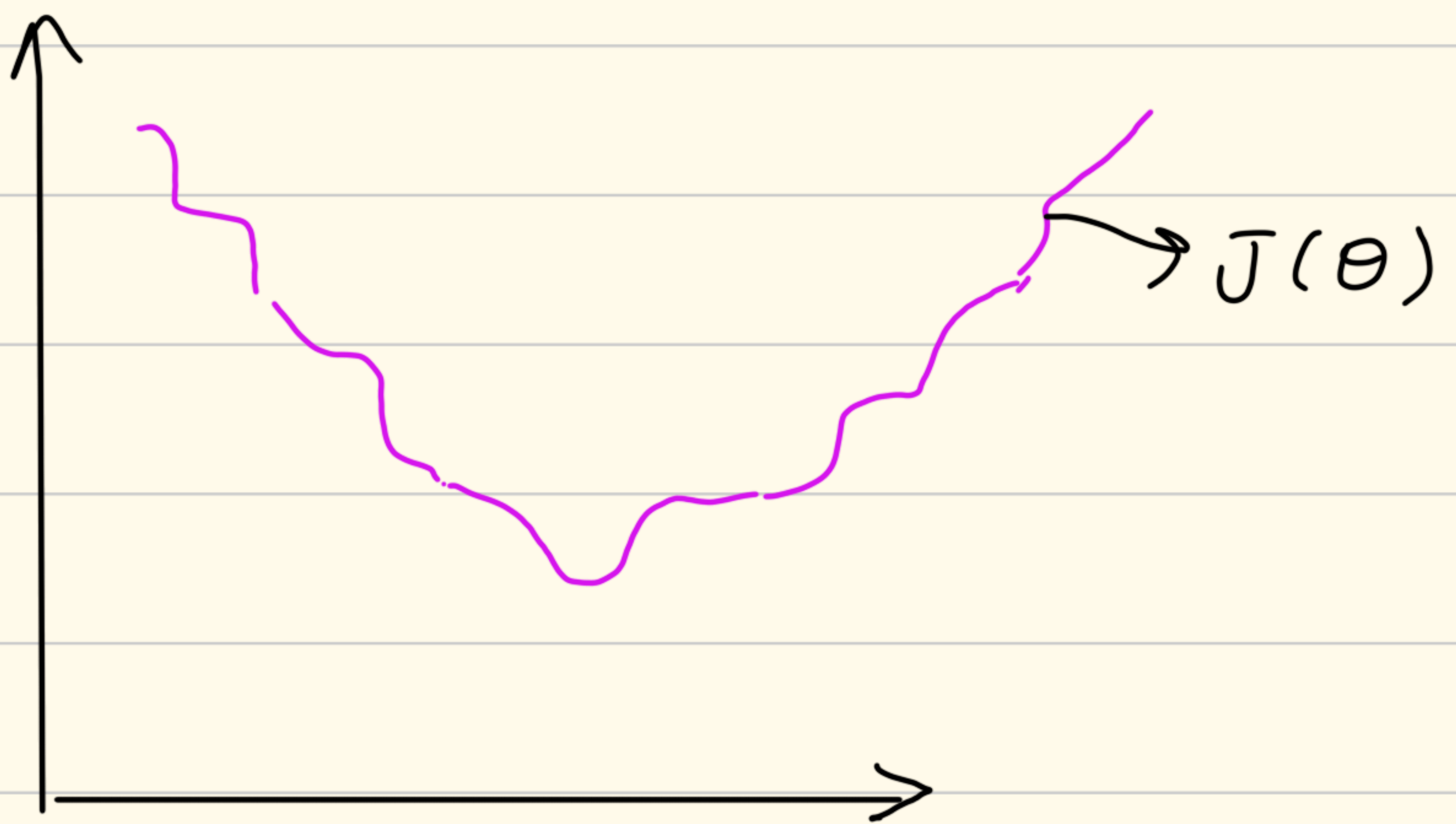


- function is convex, so we got one global minima.

→ If we map same thing in logistic regression like this:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^n \left(\frac{1}{1 + e^{-\theta^T x}} - y \right)^2$$

- This function is not convex:



so we can't use MSE here.

→ We use method of maximum likelihood here.

Maximum Likelihood Method

→ Maximum likelihood method says that "choose parameters giving maximum likelihood for a dataset."

Probability:

→ Given parameters, what is chance of observing data?

eg:

If coin is fair ($p=0.5$ here is already known parameter), what is the probability of getting 7 heads in 10 tosses?

likelihood:

→ Given observed data, how plausible are different values.

eg:

If you observed 7 heads in 10 tosses what value of p (fairness of coin) best explains this?

→ So we find parameters w such the probability of dataset given these parameters is maximum.

$$L(w) = P(\text{Data} / w)$$

↙ likelihood function

So we need maximum of this function.

Maximum Likelihood Method in Logistic Regression

→ we need to represent $P(\text{Data} / \text{weights})$.

Probability of each instance in Data can be represented using binomial representation.

means the binomial formula can give probability of 0 and 1 both classes.

$$P(y_i / x_i) = p_i^{y_i} (1 - p_i)^{1 - y_i}$$

$$\therefore p_i = \sigma(w^T x_i)$$

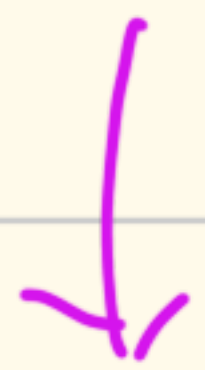
This is representation of probability of a single instance of dataset.

- if label is 1, then probability given by sigmoid would directly be probability of instance.
- If label is 0, then we do $(1 - p)$ because sigmoid give for $y=1$, for $y=0$ we do $(1 - p)$

For complete dataset:

likelihood produced by weight w :

$$L(w) = \prod_{i=1}^m p_i^{y_i} (1 - p_i)^{1 - y_i}$$



likelihood function

we need to find w that maximizes likelihood of dataset.

→ It is hard to work with multiplication, so we take log on both sides to convert it into sum(s) form.

$$\log L(w) = \sum_{i=1}^m [y_i \log p_i + (1-y_i) \log (1-p_i)]$$

log-likelihood

→ we can't maximize $\log L(w)$ easily and directly so we take its loss version and minimize that loss (effectively maximizing log likelihood)

$$\text{loss} = - \sum_{i=1}^m [y_i \log p_i + (1-y_i) \log (1-p_i)]$$

$\Rightarrow$

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y_i \log p_i + (1-y_i) \log (1-p_i) \right]$$

$$= -\frac{1}{m} \sum_{i=1}^m \left[y_i \log \overbrace{\sigma(h(x))}^{p_i} + (1-y_i) \log \left(1 - \overbrace{\sigma(h(x))}^{p_i} \right) \right]$$

we can write $\sigma(h(x))$ as $\hat{y}$

$$\Rightarrow J(\theta) =$$

$$-\frac{1}{m} \sum_{i=1}^m \left[y_i \log \hat{y} + (1-y_i) \log (1-\hat{y}) \right]$$

$$\frac{d}{d\theta_0} J(\theta) =$$

$$-\frac{1}{m} \frac{d}{d\theta_0} \sum_{i=1}^m \left[y_i \log \hat{y} + (1-y_i) \log (1-\hat{y}) \right]$$

Here chain rule would be implemented as

$$J \rightarrow \hat{y} \rightarrow h(x) \rightarrow \theta$$

$$\frac{d}{d\theta_0} J(\theta) = \underbrace{\frac{d}{d\hat{y}} J(\theta)} \times \underbrace{\frac{d}{dh(x)} \hat{y}} \times \underbrace{\frac{d}{d\theta_0} h(x)}$$

Let's find these one-by-one

(i)

$$\frac{d}{d\hat{y}} J(\theta) =$$

$$-\frac{1}{m} \frac{d}{d\hat{y}} \sum_{i=1}^m \left[y \log \hat{y} + (1-y) \log (1-\hat{y}) \right]$$

$$= -\frac{1}{m} \sum_{i=1}^m \left[y \times \frac{1}{\hat{y}} + (1-y) \times \frac{1}{(1-\hat{y})} (-1) \right]$$

$$= -\frac{1}{m} \sum_{i=1}^m \left[\frac{y(1-\hat{y}) - \hat{y}(1-y)}{\hat{y}(1-\hat{y})} \right]$$

(ii)

$$\frac{d}{dh(x)} \hat{y} = \frac{d}{dh(x)} \left(\frac{1}{1 + e^{-h(x)}} \right)$$

let's map it as $\frac{d}{dx} \left(\frac{1}{1 + e^{-x}} \right)$ and

Solve

$$\frac{d}{dx} \left(\frac{1}{1 + e^{-x}} \right) = \frac{d}{dx} (1 + e^{-x})^{-1}$$

using power rule:

$$= (-1)(1 + e^{-x})^{-2} \frac{d}{dx} (1 + e^{-x})$$

$$= (-1)(1 + e^{-x})^{-2} \times (0 + e^{-x}(-1))$$

$$= \frac{(-1)(e^{-x})(-1)}{(1+e^{-x})^2} = \frac{e^{-x}}{(1+e^{-x})^2}$$

$$= \frac{1}{(1+e^{-x})} \times \frac{e^{-x}}{(1+e^{-x})} = \sigma(x) \times \frac{e^{-x}}{(1+e^{-x})}$$

$$= \sigma(x) \times \frac{(1+e^{-x}-1)}{(1+e^{-x})}$$

$$= \sigma(x) \times \left(\frac{1+\cancel{e^{-x}}}{1+\cancel{e^{-x}}} - \frac{1}{1+e^{-x}} \right)$$

$$= \sigma(x) \times \left(1 - \frac{1}{1+e^{-x}} \right)$$

$$= \sigma(x) [1 - \sigma(x)] \rightarrow \text{derivative of sigmoid}$$

$$\Rightarrow \frac{d}{dh(x)} \sigma(h(x)) =$$

$$\sigma(h(x)) [1 - \sigma(h(x))]$$

$$= \hat{y} [1 - \hat{y}]$$

(iii)

$$\frac{d}{d\theta_0} h(x) = \frac{d}{d\theta_0} (\theta_0 + \theta_1 x_1) = 1$$

From result of (i), (ii), (iii)

$$\frac{d}{d\theta_0} J(\theta) =$$

$$= \frac{1}{m} \sum_{i=1}^m \left[\frac{y(1-\hat{y}) - \hat{y}(1-y)}{\hat{y}(1-\hat{y})} \right] \times \hat{y}(1-\hat{y})$$

$$= -\frac{1}{m} \sum_{i=1}^m [y(1-\hat{y}) - \hat{y}(1-y)]$$

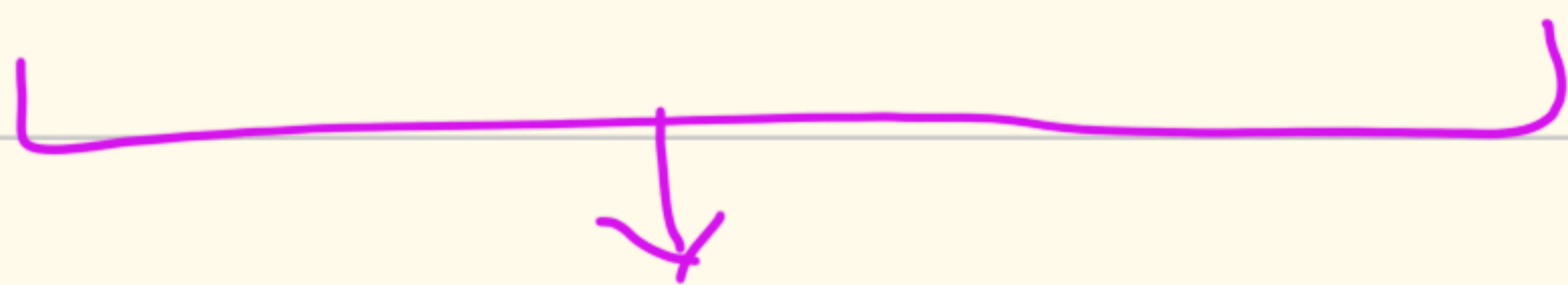
$$= -\frac{1}{m} \sum_{i=1}^m \left[y - \cancel{y\hat{y}} - \hat{y} + \cancel{y\hat{y}} \right]$$

$$= -\frac{1}{m} \sum_{i=1}^m [y - \hat{y}]$$

$$\frac{dJ(\theta)}{d\theta_0} = \frac{1}{m} \sum_{i=1}^m [\hat{y}^i - y^i]$$

$$\frac{dJ(\theta)}{d\theta_1} =$$

$$\frac{dJ(\theta)}{d\hat{y}} \times \frac{d\hat{y}}{dh(x)} \times \frac{dh(x)}{d\theta_1}$$



These would be same as

extracted for $\frac{dJ(\theta)}{d\theta_0}$ above

$$\frac{d}{d\theta_1} h(x) = \frac{d}{d\theta_1} (\theta_0 + \theta_1 x_1) = x_1$$

$\Rightarrow$

$$\frac{d}{d\theta_1} J(\theta) = \frac{1}{m} \sum_{i=1}^m [\hat{y}^i - y^i] x_1$$

General Rule:

$$\frac{d}{d\theta_j} J(\theta_1, \dots, \theta_n) =$$

$$\frac{1}{m} \sum_{i=1}^m [\hat{y}^i - y^i] x_j^i$$

Log Loss is also called Binary Cross Entropy?

Entropy:

$$\text{Entropy} = - \sum_x p(x) \log(p(x))$$

entropy = average uncertainties in system. (Here $p(x)$ is basically weight of x .)

Cross-Entropy:

true prob
distribution


$$\text{cross-entropy} = - \sum_x p(x) \log Q(x)$$

$\therefore Q(x)$ = model's prediction values

→ Cross entropy shows how

uncertain predictions are.

→ average uncertainties in model's predictions.

if we have two class and we use one-hot encoding for $p(x)$ like for $y=0$

$$p = [0, 1]$$

one-hot encoding

cross entropy =

$$- \left[\overset{y}{\uparrow} (0) \log(p(0)) + \overset{(1-y)}{\uparrow} (1) \log(p(1)) \right]$$

↓

so here it becomes same as log loss.

⇒ Log loss is also called

binary cross entropy means
log loss is special case of
cross entropy.

Binomial Logistic Regression

Example

→ we will be given with input features (x s) and output feature y (two classes as it's binomial).

x (study hours)	y (pass)
0.5	0
1	0
1.5	0
2	0
2.5	0
3	1
3.5	1
4	1
4.5	1
5	1

$$\theta_0 = -0.898$$

$$\theta_1 = 0.352$$

$$LR = 0.1$$

we need to find $\theta_{0(new)}$ and $\theta_{1(new)}$ to find

$$\sigma(h_{\theta}(x)) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

then we can use this $\sigma(h_{\theta}(x))$ to predict classes.

we know that:

$$\theta_0 = \theta_0 - (LR) \times \frac{1}{m} \sum_{i=1}^m [\hat{y}_i - y_i]$$

$$\theta_j = \theta_j - (LR) \times \frac{1}{m} \sum_{i=1}^m [\hat{y}_i - y_i] x_j^i$$

$h_{\theta}(x)$ $\theta_0 + \theta_1 x$	$\hat{y}$ $\sigma(h_{\theta}(x))$	$\hat{y} - y$	$(\hat{y} - y)x$
-0.722	0.3270	0.3270	0.1635
-0.546	0.3668	0.3668	0.3668
-0.37	0.4085	0.4085	0.61275
-0.194	0.4516	0.4516	0.9032
-0.018	0.4955	0.4955	1.2388
0.158	0.5394	-0.4606	-1.3818
0.334	0.5827	-0.4173	-1.4606
0.51	0.6248	-0.3752	-1.5008
0.686	0.6651	-0.3349	-1.5071
0.862	0.7031	-0.2969	-1.4845

(0.1504) (-4.0498)

New values of θ_0, θ_1

$$\theta_0 = (-0.898) - (0.1) \left(\frac{1}{10} \times (-4.0498) \right)$$

$$\theta_0 = -0.8965$$

$$\theta_1 = (0.352) - (0.1) \left(\frac{1}{10} \times (-4.0498) \right)$$

$$\theta_1 = 0.3925$$

$$\Rightarrow P(y=1|x) = \sigma(h_0(x))$$

$$= \frac{1}{1 + e^{-(-0.8965 + 0.3925x)}}$$

$$P(y=0|x) = 1 - P(y=1|x)$$

Multinomial Logistic

Regression Example

→ If we are given with more than two classes.

Then,

- we model a separate sigmoid for each class i.e one-vs-Rest (OVR).
- Then for an input, we check its probability in every class and predict the class giving maximum probability.

For class 0 (0 vs Rest)

C1

C2

C3

C4

OVR Class 1													
Index	x1	x2	Class	Y	h(x)	y_hat	Pred	y_hat - y	(y_hat - y)*x1	(y_hat - y)*x2	log loss		
1	1	1.2	0	1	3.6	0.9734030064	1	-0.02659699358	-0.02659699358	-0.03191639229	-0.01170731674		
2	1.3	0.8	0	1	3.8	0.9781187291	1	-0.02188127094	-0.02844565222	-0.01750501675	-0.009608425123		
3	0.8	1.5	0	1	3.4	0.9677045353	1	-0.0322954647	-0.02583637176	-0.04844319705	-0.01425722355		
4	1.5	1	0	1	3	0.9525741268	1	-0.04742587318	-0.07113880977	-0.04742587318	-0.02110121868		
5	0.7	0.9	0	1	4.8	0.9918374288	1	-0.008162571153	-0.005713799807	-0.007346314038	-0.003559506818		
6	4	4.5	1	0	-9	0.000123394576	0	0.000123394576	0.0004935783039	0.0005552755919	-0.00005359289005		
7	4.5	3.8	1	0	-8.6	0.000184071905	0	0.000184071905	0.0008283235723	0.0006994732389	-0.00007994877099		
8	3.8	4.2	1	0	-8	0.0003353501305	0	0.0003353501305	0.001274330496	0.001408470548	-0.0001456651369		
9	4.2	4	1	0	-8.4	0.0002248167702	0	0.0002248167702	0.000944230435	0.0008992670809	-0.00009764765958		
10	3.5	4.3	1	0	-7.6	0.0005002011071	0	0.0005002011071	0.001750703875	0.00215086476	-0.0002172889293		
11	7.5	1.5	2	0	-10	0.0000453978687	0	0.0000453978687	0.0003404840153	0.00006809680305	-0.00001971649141		
12	8	2	2	0	-12	0.000006144174602	0	0.000006144174602	0.00004915339682	0.0000122883492	-0.000002668389323		
13	7	1.8	2	0	-9.6	0.00006772414962	0	0.00006772414962	0.0004740690473	0.0001219034693	-0.00002941322048		
14	8.2	1.2	2	0	-10.8	0.00002039908728	0	0.00002039908728	0.0001672725157	0.00002447890474	-0.000008859301403		
15	7.8	2.5	2	0	-12.6	0.000003372003864	0	0.000003372003864	0.00002630163014	0.000008430009659	-0.00000146444514		
Q0	8				Q0 new	8.000899009	Mean	-0.008990086785	-0.01009221199	-0.009779216303	-0.00405933041		
Q1	-2				Q1 new	-1.998990779							
Q2	-2				Q2 new	-1.999022078							
LR	0.1												

$\theta_0 = 8, \theta_1 = -2, \theta_2 = -2, LR = 0.1$

Find new values:

$$\theta_0 = \theta_0 - (LR) \times \frac{1}{m} \sum_{i=1}^m \left[\overset{C1}{\hat{y}_i - y_i} \right]$$

$$= (8) - (0.1) (-0.00899)$$

$$\theta_0 = 8.000899$$

$$\theta_1 = \theta_1 - (LR) \times \frac{1}{m} \sum_{i=1}^m \left[\hat{y}_i - y_i \right] \underset{C2}{x_i}$$

$$= (-2) - (0.1) \times (-0.01009)$$

$$\theta_1 = -1.99899$$

$$\theta_2 = \theta_2 - (LR) \times \frac{1}{m} \sum_{i=1}^m [\hat{y}_i - y_i] x_2^i$$

C3

$$= (-2) - (0.1) \times (-0.00977)$$

$$\theta_2 = -1.999023$$

$\Rightarrow$

$$\sigma(h_\theta(x)) =$$

|

$$- (8.000899 - 1.99899x_1 - 1.999023x_2)$$

1 + e

For class 1 (1 vs Rest)

OVR Class 2 											
	x1 	x2 	Class 	Y 	h(x) 	y_hat 	Pred 	y_hat - y 	(y_hat - y)*x1 	(y_hat - y)*x2 	log loss 
1	1	1.2	0	0	-12.32	00000446159397	0	0.000004461593978	0.000004461593978	0.000005353912774	-0.000001937649968
2	1.3	0.8	0	0	-13.58	00000126555314	0	0.000001265553141	0.000001645219084	0.000001012442513	-0.0000005496230936
3	0.8	1.5	0	0	-11.35	.0000117693511	0	0.00001176935111	0.000009415480885	0.00001765402666	-0.00000511139432
4	1.5	1	0	0	-12.6	00000337200386	0	0.000003372003864	0.000005058005796	0.000003372003864	-0.00000146444514
5	0.7	0.9	0	0	-13.79	00000102583768	0	0.000001025837685	0.0000007180863792	0.0000009232539162	-0.0000004455158743
6	4	4.5	1	1	3.55	0.9720774262	1	-0.02792257378	-0.1116902951	-0.125651582	-0.01229914203
7	4.5	3.8	1	1	1.32	0.7891817065	1	-0.2108182935	-0.9486823206	-0.8011095152	-0.1028229904
8	3.8	4.2	1	1	2.18	0.8984390721	1	-0.1015609279	-0.385931526	-0.4265558972	-0.04651136936
9	4.2	4	1	1	1.8	0.8581489351	1	-0.1418510649	-0.5957744726	-0.5674042596	-0.06643733211
10	3.5	4.3	1	1	2.27	0.9063617878	1	-0.09363821224	-0.3277337428	-0.4026443126	-0.04269841263
11	7.5	1.5	2	0	-4.65	0.009471043582	0	0.009471043582	0.07103282686	0.01420656537	-0.004132824084
12	8	2	2	0	-2.2	0.09975048912	0	0.09975048912	0.798003913	0.1995009782	-0.04563710592
13	7	1.8	2	0	-3.98	0.0183428907	0	0.0183428907	0.1284002349	0.03301720326	-0.008040183845
14	8.2	1.2	2	0	-5.12	0.005940522198	0	0.005940522198	0.04871228203	0.007128626638	-0.002587629578
15	7.8	2.5	2	0	-0.45	0.3893607661	0	0.3893607661	3.037013975	0.9734019151	-0.2142152955
Q0	-18				Q0 new	-17.99964731	Mean	-0.003526897754	0.1142248115	-0.07307213082	-0.03635945293
Q1	1				Q1 new	0.9885775188					
Q2	3.9				Q2 new	3.907307213					
LR	0.1										

$\sigma(h_{\theta}(x)) =$

1

$-(-17.9965 + 0.98857x_1 + 3.9073(x_2)$

$1 + e$

For class 2 (2 vs Rest)

OVR Class 3												
Index	x1	x2	Class	Y	h(x)	y_hat	Pred	y_hat - y	(y_hat - y)*x1	(y_hat - y)*x2	log loss	
1	1	1.2	0	0	-9.2	0.0001010291939	0	0.0001010291939	0.0001010291939	0.0001212350327	-0.00004387863797	
2	1.3	0.8	0	0	-7.9	0.0003706061406	0	0.0003706061406	0.0004817879828	0.0002964849125	-0.0001609820341	
3	0.8	1.5	0	0	-10.1	0.00004107786776	0	0.00004107786776	0.00003286229421	0.00006161680165	-0.00001784025772	
4	1.5	1	0	0	-7.5	0.0005527786369	0	0.0005527786369	0.0008291679554	0.0005527786369	-0.0002401350886	
5	0.7	0.9	0	0	-9.8	0.00005544852472	0	0.00005544852472	0.00003881396731	0.00004990367225	-0.00002408165597	
6	4	4.5	1	0	-3.5	0.02931223075	0	0.02931223075	0.117248923	0.1319050384	-0.01292044249	
7	4.5	3.8	1	0	-1.3	0.214165017	0	0.214165017	0.9637425763	0.8138270644	-0.1046686416	
8	3.8	4.2	1	0	-3.8	0.02188127094	0	0.02188127094	0.08314882956	0.09190133793	-0.009608425123	
9	4.2	4	1	0	-2.4	0.08317269649	0	0.08317269649	0.3493253253	0.332690786	-0.03771246171	
10	3.5	4.3	1	0	-4.8	0.008162571153	0	0.008162571153	0.02856899904	0.03509905596	-0.003559506818	
11	7.5	1.5	2	1	10	0.9999546021	1	-0.0000453978687	-0.0003404840153	-0.00006809680305	-0.00001971649141	
12	8	2	2	1	11	0.9999832986	1	-0.00001670142185	-0.0001336113748	-0.0000334028437	-0.00000725339592	
13	7	1.8	2	1	8.2	0.9997254218	1	-0.0002745781561	-0.001922047093	-0.000494240681	-0.0001192641525	
14	8.2	1.2	2	1	12.4	0.9999958814	1	-0.000004118571745	-0.00003377228831	-0.000004942286094	-0.000001788676666	
15	7.8	2.5	2	1	9.9	0.9999498278	1	-0.00005017216468	-0.0003913428845	-0.0001254304117	-0.0000217900409	
Q0	-11				Q0 new	-11.00238283	Mean	0.02382825056	0.1027131371	0.09371861258	-0.01127508054	
Q1	3				Q1 new	2.989728686						
Q2	-1				Q2 new	-1.009371861						
LR	0.1											

$\sigma(h_0(x)) =$

|

$-(-11.00238 + 2.98973x_1 - 1.00937x_2)$

$1 + e$

Softmax Method for Multinomial Case

→ It is another method of finding probabilities in the multinomial case.

$$p(y=k|x) = \frac{e^{z_k}}{\sum_{i=1}^n e^{z_i}}$$

predict class with max probability.

Example:

Suppose we have following

values for raw scores

$$z_1 = -2x_1 - 2x_2 + 8$$

$$z_2 = x_1 + 3.9x_2 - 18$$

$$z_3 = 3x_1 - x_2 - 11$$

we need to check point (1.2, 1.1)

$$\begin{aligned} z_1 &= -2(1.2) - 2(1.1) + 8 \\ &= 3.4 \end{aligned}$$

$$\begin{aligned} z_2 &= (1.2) + 3.9(1.1) - 18 \\ &= -12.51 \end{aligned}$$

$$\begin{aligned} z_3 &= 3(1.2) - (1.1) - 11 \\ &= -8.5 \end{aligned}$$

Applying Softmax:

$$P_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} = 0.9999$$

$$P_2 = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}} = 0.0000009$$

$$P_3 = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}} = 0.0000067$$

As P_1 is highest, so predicted class is 1.